

Question-3

Using crt.sh to get the subdomains of “iiitd.edu.in”

- For this, I simply went to the crt.sh and, searched for the subdomain, and entered “%.iiitd.edu.in” in the search box. Here, “%” is a wildcard character, which will ensure we get results that match the subdomain.
- Below is the ss for the same.

crt.sh Certificate Search

Enter an **Identity** (Domain Name, Organization Name, etc),
a **Certificate Fingerprint** (SHA-1 or SHA-256) or a **crt.sh ID**:

%.iiitd.edu.in

Search [Advanced...](#)

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- After this, I got a list of subdomains under iiitd.edu.in. And I downloaded this in json format for future use.

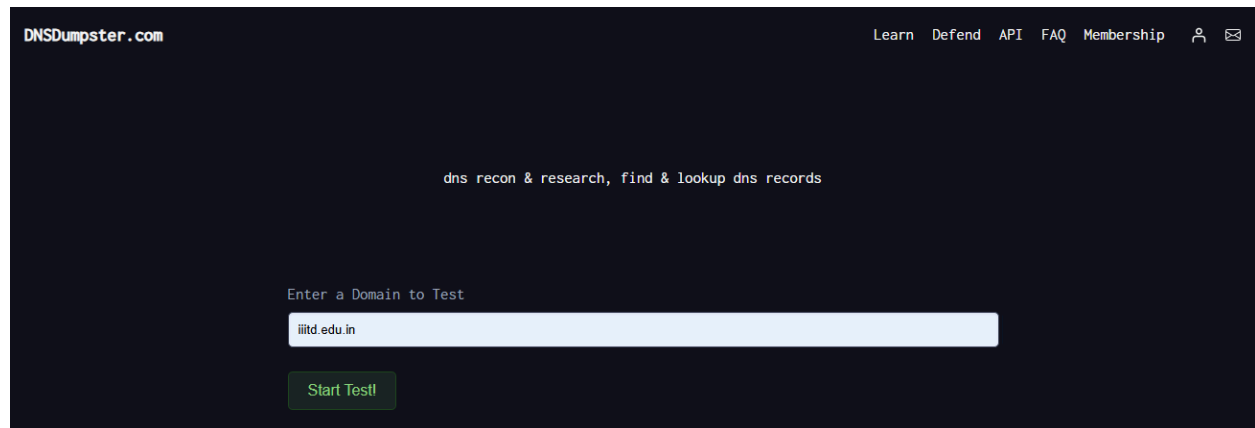
crt.sh Identity Search    [Group by Issue](#)

Criteria Type: Identity Match: ILIKE Search: 'iiitd.edu.in'

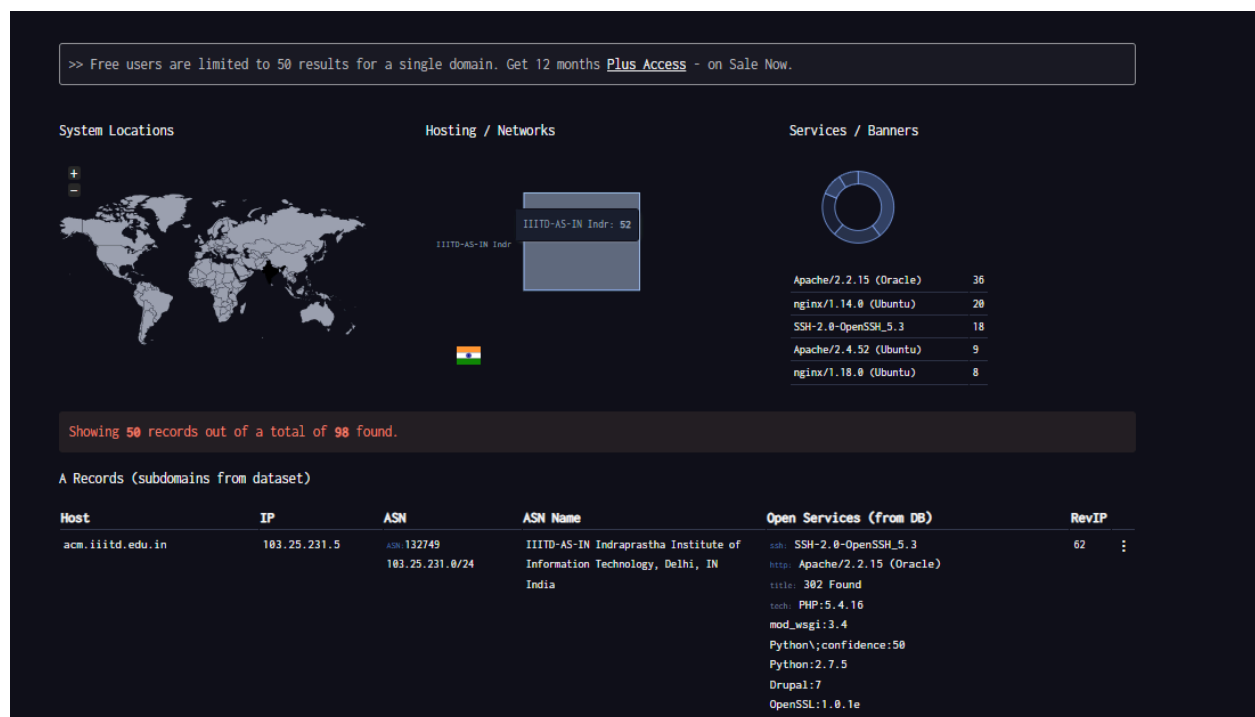
Certificates	crt.sh ID	Logged At	Not Before	Not After	Common Name	Matching Identities	Issuer Name
	16764491490	2025-02-16	2025-02-16	2025-05-17	one-ird.iiitd.edu.in	one-ird.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	16735858479	2025-02-15	2025-02-15	2025-05-16	ims.iiitd.edu.in	ims.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R11
	16736133923	2025-02-14	2025-02-14	2025-05-15	webs.iiitd.edu.in	webs.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	16736148707	2025-02-14	2025-02-14	2025-05-15	webs.iiitd.edu.in	webs.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	16736230818	2025-02-14	2025-02-14	2025-05-15	webs.iiitd.edu.in	webs.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R11
	16743964717	2025-02-14	2025-02-14	2025-05-15	webs.iiitd.edu.in	webs.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	1668246610	2025-02-11	2025-02-11	2025-05-12	federatedhealthplatform.taviab.iiitd.edu.in	federatedhealthplatform.taviab.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	16657492010	2025-02-10	2025-02-10	2025-05-11	arms.iiitd.edu.in	arms.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	16452015962	2025-02-10	2025-02-10	2025-05-11	idp.iiitd.edu.in	idp.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	16641018296	2025-02-09	2025-02-09	2025-05-10	claps.iiitd.edu.in	claps.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	16557418803	2025-02-06	2025-02-06	2025-05-07	chikitsachakra.taviab.iiitd.edu.in	chikitsachakra.taviab.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	16568789673	2025-02-06	2025-02-06	2025-05-07	chikitsachakra.taviab.iiitd.edu.in	chikitsachakra.taviab.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	16466690187	2025-02-06	2025-02-06	2025-05-07	api.irs.iiitd.edu.in	api.irs.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	16583946521	2025-02-04	2025-02-04	2025-05-05	sapnansl.nri.iiitd.edu.in	sapnansl.nri.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R11
	16388283119	2025-02-03	2025-02-03	2025-05-04	greenpreneurs.iiitd.edu.in	greenpreneurs.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R6
	16486632793	2025-02-01	2025-02-01	2025-05-02	taviab.iiitd.edu.in	taviab.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R11
	16539347822	2025-02-01	2025-02-01	2025-05-02	taviab.iiitd.edu.in	taviab.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R11
	16476476440	2025-01-31	2025-01-31	2025-05-01	raylab.iiitd.edu.in	raylab.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	16491694288	2025-01-31	2025-01-31	2025-05-01	booking.iiitd.edu.in	booking.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	16521499345	2025-01-31	2025-01-31	2025-05-01	raylab.iiitd.edu.in	raylab.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	16616458885	2025-01-30	2025-01-30	2025-04-30	nodes.iiitd.edu.in	nodes.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	16473229883	2025-01-29	2025-01-29	2025-04-29	importal.iiitd.edu.in	importal.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	16449214120	2025-01-28	2025-01-28	2025-04-28	crohnadsadmin.taviab.iiitd.edu.in	crohnadsadmin.taviab.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R6
	16459168772	2025-01-28	2025-01-28	2025-04-28	crohnadsapi.taviab.iiitd.edu.in	crohnadsapi.taviab.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R6
	16450104347	2025-01-28	2025-01-28	2025-04-28	crohnadsadmin.taviab.iiitd.edu.in	crohnadsadmin.taviab.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R6
	16488204024	2025-01-28	2025-01-28	2025-04-28	crohnads.taviab.iiitd.edu.in	crohnads.taviab.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R6
	16727974164	2025-01-24	2025-01-24	2025-04-24	ims.ecelabs.iiitd.edu.in	ims.ecelabs.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	16396918725	2025-01-24	2025-01-24	2025-04-24	ims.ecelabs.iiitd.edu.in	ims.ecelabs.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R10
	16392665055	2025-01-23	2025-01-23	2025-04-23	taallocation.iiitd.edu.in	taallocation.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R11
	16445458708	2025-01-23	2025-01-23	2025-04-23	taallocation.iiitd.edu.in	taallocation.iiitd.edu.in	C=US, O=Let's Encrypt, CN=R11

Using dnsdumpster to get the subdomains of “iiitd.edu.in”:-

- For this, I repeated the same thing I did with crt.sh, except the input query didn’t have any wildcard character, as this is not supported.



- Below are the results:-



- After this, I saved this in the JSON format, too, by manually requesting this page via postman API.

Part -A: Fetching the private IP address of these domains:-

- To fetch the private IP address of the subdomains I got in the previous section, I used a simple program in JavaScript, which uses the inbuilt dns library to resolve the subdomains.
- To achieve this, I first parsed the JSON data I saved in the earlier section, and then inside a simple for loop, I used the resolve4Sync function provided in the dns library and passed the parsed subdomain name.
- I repeated this for both the JSON files of crt.sh and dnsdumpster.
- After getting the IP addresses, I saved them to a CSV file "DOMAIN_TO_IP.csv."
- Below is the sample output for this in the format as given in the assignment : (subdomain:[PRIVATE_IP]).

```
PS C:\Users\DELL\Desktop\FCS-Assignments\Assignment-1> node "c:\Users\DELL\Desktop\FCS-Assignments\Assignment-1\q3.js"
Domain name and IP address saved to DOMAIN_TO_IP.csv
Part A Answer:-
Domain Name : IP
one-ird.iiitd.edu.in : 192.168.3.81
rms.iiitd.edu.in : 192.168.3.172
webs.iiitd.edu.in : 192.168.22.132
webs.iiitd.edu.in : 192.168.22.132
webs.iiitd.edu.in : 192.168.22.132
webs.iiitd.edu.in : 192.168.22.132
federatedhealthplatform.tavlab.iiitd.edu.in : 192.168.3.144
arms.iiitd.edu.in : 192.168.1.79
idp.iiitd.edu.in : 192.168.1.31
claps.iiitd.edu.in : 192.168.3.56
chikitsachakra.tavlab.iiitd.edu.in : 192.168.3.74
chikitsachakra.tavlab.iiitd.edu.in : 192.168.3.74
api.irb.iiitd.edu.in : 192.168.3.128
sapnansl.nrl.iiitd.edu.in : 192.168.29.160
greenpreneurs.iiitd.edu.in : 46.202.166.45
tavlab.iiitd.edu.in : 192.168.3.74
tavlab.iiitd.edu.in : 192.168.3.74
raylab.iiitd.edu.in : 192.168.30.176
```

Part -B: Automation of this process:-

- To automate this task, I wrote a program in JavaScript that uses the fetch API of JavaScript to perform a get request for the crt.sh and dnsdumpster.
- First, I defined the static URL for both of the sites, which are:-
 1. Crt.sh: `CRTSH_URL = "https://crt.sh/?q="`
 2. Dnsdumpster: `DNSDUMPSTER_URL = "https://api.dnsdumpster.com/domain/"`
- Next, I hit the desired URL by passing the search query (the subdomain for which you want to search). After this, I checked whether the responses from the servers were okay (status codes), and then, if they were okay, I parsed the JSON data I got from the site's servers.
- After getting the JSON data, I reused the functions I wrote in part-a, to get the IP address from the subdomains and then saved the results to the CSV files, too.
- Also, dnsdumpster had an official API, so I leveraged it by putting my generated API key in the request header.
- These two tasks (each one for each site) happened asynchronously (parallel execution), which also increased the speed of the lookups!
- To change the search query, simply change the value of `const SEARCH_QUERY = "iiitd.edu.in"`
- Below is the output for the same (CSV files are attached):-

```
PS C:\Users\DELL\Desktop\FCS-Assignments\Assignment-1> node "c:\Users\DELL\Desktop\FCS-Assignments\Assignment-1\q3.js"
Automation of DNSDUMPSTER DONE!!
Domain name and IP address saved to AUTOMATED_DNSDUMPSTER_DOMAIN_TO_IP.csv
Automation of CRT.SH DONE!!
Domain name and IP address saved to AUTOMATED_CRTSH_DOMAIN_TO_IP.csv
```

Part -C: Implications of Private Addresses being leaked-

- If this information about the private addresses gets leaked to the public, then the attacker can't directly attack the network, as these are private addresses and can only be used in the network within IIITD, so from an attacker's pov this information still would not help indirectly interfering with the network.
- However, an attacker can use this information to gather insider details about the internal network architecture of the IIITD.
- And thus use this information for reconnaissance or understanding the network of the IIITD better.
- And further, this information can be used for social engineering & phishing attacks, if an attacker has an idea of the internal network, then he can craft specific payloads for these phishing attacks.
- Furthermore, if any of these IP addresses don't have proper configurations set and have misconfigured firewalls, then attacker could exploit this vulnerability.

References

- crt.sh: <https://crt.sh/>
- Dnsdumpster: <https://dnsdumpster.com/>
- Dnsdumpster for developers: <https://dnsdumpster.com/developer/>
- Fetch API in JS:
https://developer.mozilla.org/en-US/docs/Web/API/Fetch_API